

Magnetization evolution in connected pore systems

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The equations governing magnetization evolution in fluid-filled pore systems are developed. Particular attention is given to the effect of coupling between pores. In pore systems with a range of pore sizes (and/or decay rates), the evolution of the total dipole moment is described exactly by a spectrum of decay rates, leading to multiple-exponential decay. We study this spectrum of decay rates as a function of coupling strength between pores, using perturbation theory, effective-medium theory, and matrix diagonalization. The spectrum of decay rates evolves from the individual pore decay-rate distribution, at zero coupling, to a δ -function distribution, at infinite coupling. The development of the spectrum of decay rates between the two extremes of weak and strong coupling is discussed. The three methods employed to study this problem are in good agreement. The effect of coupling between pores is important in NMR measurements when comparing different porous materials and in the study of the temperature dependence of characteristic decay rates in these materials.

I. INTRODUCTION

Porous-media systems consist of matrix, pore fluid, and probe particle. For example, in water saturated fused glass beads, the glass matrix is complementary to the pore space which is filled with weakly ionized water. In an electrical measurement the probe particle is an ion, e.g., Na^+ , which probes the pore space as a participant in the current carrying process. In an NMR measurement, the probe particle is the H atom on a water molecule, the carrier of a H nuclear magnetic moment. This molecule explores the pore space by diffusive motion. The magnetic moment it carries relaxes due to dipolar interactions with other molecules and interactions at the pore-fluid-matrix interface. The magnetization evolves by diffusive motion and relaxation processes. It responds to features of the geometry of the pore space. We are interested in the geometrical information which may be garnered from NMR studies of magnetization evolution.

Brownstein and Tarr,¹ in a classic work, have discussed magnetization evolution in independent pores. Our interest is in magnetization evolution in coupled-pore systems. We take the characteristics of the diffusive motion and the relaxation mechanisms to be known and concern ourselves with geometry. The geometry of the pore system we wish to study is that of a lattice of pores (e.g., sites characterized by volume and surface) connected by throats (e.g., bonds characterized by cross-sectional area). See Fig. 1. Thus, we describe the pore space with a set of geometrical parameters: V the volume of a pore; A the surface area of a pore; S the cross-sectional area of a

throat; and l the distance between the center of adjacent pores. In addition, we define a pore size a , where $a = V/A$.

The behavior of magnetization in a pore space is controlled by four rates.

(1) $w_B = 1/T_{1,B}$, the bulk relaxation rate, an intrinsic property of the pore fluid.

(2) $w_p = \rho/a$, the relaxation rate in the vicinity of the fluid-matrix interface. [$\rho = d(1/\tau_w - 1/T_{1,B})$], where d is the thickness of the fluid layer next to the matrix walls

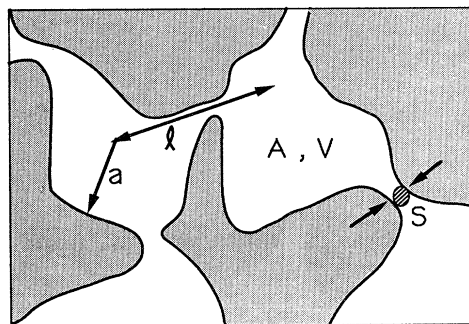


FIG. 1. Schematic of a porous-medium cross-sectional area. The porous medium is characterized by large voids, each assigned a volume V and surface area A , connected by narrow channels of cross-sectional area S . The distance between pores is l and $a = V/A$.

which is influenced by the matrix and τ_w^{-1} is the relaxation rate of fluid in d . Thus, ρ has units of a velocity and characterizes how well the matrix relaxes the fluid in contact with it.]

(3) $w_D = D/a^2$, the rate of molecular diffusion to the fluid-matrix interface (D is the bulk fluid diffusion coefficient).

(4) $w_c = SD/Vl$, the rate of molecular diffusion between pores, the inter pore coupling rate.

The effect of a finite value for w_B can be accounted for exactly² and simply in terms of the solution when $w_B \equiv 0$, thus we will neglect w_B in our discussion. Also, note $w_c < w_D$.

The work of Brownstein and Tarr¹ (BT) on single pores corresponds to the weak coupling limit $w_c \rightarrow 0$. These authors describe the behavior of the magnetization in the slow diffusion limit $w_p \gg 10w_D$, the crossover from slow diffusion to fast diffusion at $w_p \approx w_D$, and the behavior of the magnetization in the fast diffusion limit $w_p \ll w_D$. See Fig. 2. In the fast diffusion limit, the surface relaxation rate is slow compared to the rate of diffusive magnetization equilibration; the magnetization is approximately uniform across a pore and the decay is single exponential. Coupling between pores can be important in the fast diffusion regime.

In Fig. 2 we show that BT have discussed the behavior of the magnetization in going from point 1 to 2 in rate space. In this paper we develop methods of studying the behavior of the magnetization in going from point 2 to point 3 in this space. The BT methodology is valid here, in principle, but cumbersome in describing this evolution, in practice. It seems to be the case²⁻⁴ that sedimentary rocks correspond to a point somewhere along the 2→3 line of Fig. 2. Individual pores would be in the fast diffusion limit were they uncoupled from their neighbors and the spectrum of decay rates is conventionally interpreted as a spectrum of pore sizes. If the coupling w_c is "strong enough," however, then the entire pore space becomes a single pore in the fast diffusion limit and the de-

cay spectrum narrows to a δ function at the rate $w_p = \rho A/V$, where A and V are the total area and pore volume of the system. Decay in this limit is strictly single exponential, $M(t) = M(0)e^{-w_p t}$, where M is the total dipole moment of the system. A goal of this paper is to understand this evolution and what constitutes "strong enough" coupling. Since w_c is proportional to the diffusion constant, it can be increased by a factor of ~ 20 simply by superheating the sample past the boiling point of water. Thus, we feel that temperature dependent decay spectra should prove to be a powerful probe of the physics of magnetization evolution.

Section II contains a description of various measures of magnetic behavior that concern us. The model porous medium we study is described. The equations of motion of the magnetization in this porous medium are discussed. In Sec. III, we develop perturbation theory solutions to the equations of motion in both the weak and strong coupling limits (points 2 and 3 in Fig. 2). Section IV contains a description and results of an effective medium theory treatment of the equations of motion. We discuss numerically exact solutions to the equation of motion for several small system realizations of the model porous medium in Sec. V. Finally, in Sec. VI we compare the three methods we have employed to study the behavior of magnetization and discuss the relevance of our results to physical systems.

II. MAGNETIZATION EQUATIONS AND POROUS MEDIA MODEL

In a typical magnetization measurement, the time evolution of the total dipole moment is probed, i.e., $M(t)$ versus t . The total dipole moment may decay as a single exponential, a multiple exponential,³ a stretched exponential,² etc. To examine the content of a decay curve in a model independent way, we define the mean lifetime:

$$\tau_m = \frac{\int_0^\infty t (dM/dt) dt}{\int_0^\infty (dM/dt) dt} = \frac{\int_0^\infty M(t) dt}{M(0)}, \quad (1)$$

where $M(t)$ is the total dipole moment of the system at time t . Note that τ_m is directly related to the Fourier-Laplace transform of M , $\tilde{M}(s, k)$,

$$\tau_m = \frac{1}{M(0)} \lim_{s \rightarrow 0} \lim_{k \rightarrow 0} \tilde{M}(s, k). \quad (2)$$

This quantity is discussed by Torquato⁵ in the context of permeability and by Wilkinson, Johnson, and Schwartz⁶ in the context of magnetization decay in isolated pores. Given that the magnetization evolution involves both diffusion and decay, it is reasonable to expect to represent (M) in the form

$$\frac{M(t)}{M(0)} = \int_0^\infty d\lambda n(\lambda) e^{-\lambda t}, \quad (3)$$

where $n(\lambda)$ is the spectrum of decay rates. In fact $n(\lambda)$ can be expressed uniquely as $n(\lambda) = \sum_n I_n \delta(\lambda - \lambda_n)$, where λ_n is the eigenvalue and I_n its amplitude, as discussed by BT. Then,

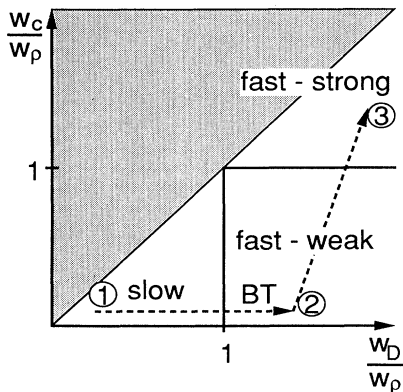


FIG. 2. Rate space for magnetization decay in pore systems. Brownstein and Tarr discuss evolution from slow to fast diffusion, 1→2. We discuss evolution from weak to strong coupling, 2→3.

$$\tau_m = \int_0^\infty d\lambda \frac{1}{\lambda} n(\lambda) \equiv \langle \lambda^{-1} \rangle, \quad (4)$$

i.e., τ_m is the appropriately weighted average of λ^{-1} . Thus, τ_m , the $n = -1$ moment of $n(\lambda)$, favors small λ or slow decay rates.

At early time, $t \rightarrow 0$, the decay of the total dipole moment is given by

$$\frac{M(t)}{M(0)} = 1 - \frac{t}{\tau_0}, \quad (5)$$

where

$$\frac{1}{\tau_0} = \int_0^\infty d\lambda \lambda n(\lambda) \equiv \langle \lambda \rangle. \quad (6)$$

The short-time behavior provides a measure of the $n = 1$ moment of $n(\lambda)$. Given that surface effects are the dominant mechanism for relaxation, magnetization decay at the very shortest times is proportional to the total matrix surface to total fluid volume in the system;⁷ i.e., $1/\tau_0 = \rho A/V = w_\rho$.

In the strong coupling limit, e.g., $D \rightarrow \infty$, the entire system can be viewed as a single pore with total dipole moment decay given by its surface to volume ratio; i.e., $1/\tau_m = \rho A/V$. Thus,

$$\lim_{D \rightarrow \infty} \frac{1}{\tau_m} = \frac{1}{\tau_0}. \quad (7)$$

We will focus our attention on the quantities τ_m , τ_0 , and $n(\lambda)$. Through calculations of $M(t)$ in terms of $n(\lambda)$ we hope to learn $n(\lambda)$ (and therefore τ_m and τ_0) and its relationship to the geometry of the pore space.

While the fluid-saturated rocks in which we are primarily interested do not contain a regular lattice of pores, they do contain a multiply connected system of pores far from the percolation threshold. In addition, the magnetization evolution is governed by diffusive processes which are independent of the precise connectivity of the lattice. Thus, to have a specific model of a pore system, we choose a hypercubic lattice of pores and throats. We identify pores with sites in a lattice and throats with bonds between sites. Each site is characterized by a volume V_i and a surface area A_i . Each bond is characterized by a cross-sectional area S_{ij} and a length l_{ij} . Thus, in a hypercubic lattice model in d dimension, we admit N possible values of V and A , and dN possible values of l and S , where N is the number of sites. For example, in a particularly simple model we might take $A_i = 4\pi R_i^2$, $V_i = 4\pi R_i^3/3$, $l_{ij} = \bar{l}$, and $S_{ij} = \bar{S}$, where R_i is a pore radius.

Following Cohen and Mendelson,⁴ we assume that each pore is in the fast diffusion limit (or would be if it were uncoupled from the rest of the system) and that all of the lengths l_{ij} are the same. We are led to Eq. (6.15a) of Cohen and Mendelson⁴ for the dipole moment in pore i ,

$$\frac{dM_i}{dt} = -w_\rho r_i M_i + w_c \sum_j^{NN} s_{ij} \left[\frac{M_j}{v_j} - \frac{M_i}{v_i} \right] \equiv - \sum_j \mathcal{B}_{ij} M_j, \quad (8)$$

where $r_i = A_i \bar{V} / \bar{A} V_i$, $s_{ij} = S_{ij} / \bar{S}$, $v_i = V_i / \bar{V}$, $w_c = D \bar{S} / l \bar{V}$, $w_\rho = \rho \bar{A} / \bar{V}$ (note, $w_\rho \equiv 1/\tau_0$) and the sum is over all sites j which are nearest neighbors (NN) to i . Overbars indicate averages.

We seek solutions to Eq. (8) subject to the initial condition at $t=0$ of uniform magnetization, $M_i(0) = \bar{m} V_i$, where $\bar{m}$ is the magnetization. Since $s_{ij} = s_{ji}$, but $v_i \neq v_j$ (usually) the operator $\mathcal{B}_{ij}$ is not symmetric in the interchange of i and j . Equation (8) is symmetrized by using $M_i = \sqrt{v_i} \theta_i$; then

$$\frac{d\theta_i}{dt} = -\mathcal{H}_{ij} \theta_j, \quad (9)$$

where

$$\mathcal{H}_{ij} = \left[w_\rho r_i + w_c \sum_k^{NN} \frac{s_{ik}}{v_i} \right] \delta_{ij} - w_c \frac{s_{ij}}{\sqrt{v_i v_j}}, \quad (10)$$

and $s_{ij} = 0$ for j not a nearest neighbor of i . Given our initial condition we find that the total dipole moment $M(t) = \sum_i M_i$ can be expressed as

$$\frac{M(t)}{M(0)} = \langle \theta(0) | \theta(t) \rangle, \quad (11)$$

where

$$|\theta(0)\rangle = \frac{1}{\sqrt{N}} \begin{bmatrix} \sqrt{v_1} \\ \sqrt{v_2} \\ \vdots \\ \sqrt{v_N} \end{bmatrix} \quad (12)$$

and $\langle \theta(0) | \theta(0) \rangle = 1$. Expanding $|\theta\rangle$ in terms of the complete set of eigenstates of $\mathcal{H}$,

$$|\theta(t)\rangle = \sum_n a_n(t) |\psi_n\rangle, \quad (13)$$

where

$$\mathcal{H} |\psi_n\rangle = \lambda_n |\psi_n\rangle \quad (14)$$

[in the present case the matrix, Eq. (10), is real symmetric, the $|\psi_n\rangle$ are a complete set of orthonormal states and the λ_n are positive definite] we find

$$|\theta(t)\rangle = \sum_n \langle \psi_n | \theta(0) \rangle e^{-\lambda_n t} |\psi_n\rangle \quad (15)$$

and

$$\frac{M(t)}{M(0)} = \sum_n |\langle \theta(0) | \psi_n \rangle|^2 e^{-\lambda_n t}. \quad (16)$$

Upon comparison with Eq. (3) we have

$$n(\lambda) = \sum_n |\langle \theta(0) | \psi_n \rangle|^2 \delta(\lambda - \lambda_n). \quad (17)$$

The rates λ_n are the eigenvalues of $\mathcal{H}$. The amplitude at each rate λ_n is the overlap of $|\psi_n\rangle$, from Eq. (14), with $\langle \theta(0) |$, the state created in experiment by making the magnetization uniform at $t=0$. In the next three sections of this paper we will describe three methods for developing solutions to Eqs. (16) and (17).

III. PERTURBATION THEORY FOR STRONG AND WEAK COUPLINGS

We wish to find perturbation theory solutions to Eq. (9), given the Hamiltonian in Eq. (10). The ratio w_c/w_ρ determines whether the system is in the weak coupling limit (near point 2 in Fig. 2) or the strong coupling limit (near point 3 in Fig. 2). Therefore, we rewrite Eq. (9) as

$$\frac{d|\theta\rangle}{d\tau} = -(\mathcal{H}^{(0)} + \epsilon\mathcal{V})|\theta\rangle, \quad (18)$$

where ϵ is the small parameter and the quantities τ , $\mathcal{H}^{(0)}$, and $\mathcal{V}$ are chosen according to the region of interest.

Strong coupling:

$$\begin{aligned} \epsilon &= \frac{w_\rho}{w_c}, \quad \tau = w_c t, \\ \mathcal{H}_{ij}^{(0)} &= \sum_k \frac{s_{ik}}{v_i} \delta_{ij} - \frac{s_{ij}}{\sqrt{v_i v_j}}, \\ \mathcal{V}_{ij} &= r_i \delta_{ij}. \end{aligned} \quad (19)$$

Weak coupling:

$$\begin{aligned} \epsilon &= \frac{w_c}{w_\rho}, \quad \tau = w_\rho t, \quad \mathcal{H}_{ij}^{(0)} = r_i \delta_{ij}, \\ \mathcal{V}_{ij} &= \sum_k \frac{s_{ik}}{v_i} \delta_{ij} - \frac{s_{ij}}{\sqrt{v_i v_j}}. \end{aligned} \quad (20)$$

Using standard Schrödinger perturbation theory⁸ we may investigate either the strong or the weak coupling limit. We assume we know the eigenvalues E_n and eigenkets $|n\rangle$ which satisfy

$$\mathcal{H}^{(0)}|n\rangle = E_n|n\rangle, \quad (21)$$

and look for solutions to

$$(\mathcal{H}^{(0)} + \epsilon\mathcal{V})|\psi_n\rangle = \lambda_n|\psi_n\rangle \quad (22)$$

in terms of E_n and $|n\rangle$. To second order in the perturbation,

$$\lambda_n = E_n + \epsilon\langle n|\mathcal{V}|n\rangle + \epsilon^2 \sum_{k(\neq n)} \frac{|\langle k|\mathcal{V}|n\rangle|^2}{E_n - E_k}, \quad (23)$$

and to first order

$$|\psi_n\rangle = c_n \left[|n\rangle + \epsilon \sum_{k(\neq n)} |k\rangle \frac{\langle k|\mathcal{V}|n\rangle}{E_n - E_k} \right] \quad (24)$$

Here c_n is given by

$$c_n = 1 - \frac{\epsilon^2}{2} \sum_{k(\neq n)} \frac{|\langle k|\mathcal{V}|n\rangle|^2}{(E_n - E_k)^2}, \quad (25)$$

correct to second order in the perturbation. From Eq. (16) the normalized total dipole moment is given by

$$\frac{M(t)}{M(0)} = - \sum_n |\langle \theta(0)|\psi_n\rangle|^2 e^{-\lambda_n \tau}. \quad (26)$$

The problem at hand is to solve Eq. (21) for $|n\rangle$ and E_n and to use the results to find $M(t)$.

We consider the strong coupling case first. By inspection we see that the ground state of $\mathcal{H}^{(0)}$, with energy $E_0=0$, is equivalent to the $t=0$ solution for $|\theta\rangle$ described in Sec. II, i.e., $|0\rangle = |\theta(0)\rangle$, where $|\theta(0)\rangle$ is given in Eq. (12). Since $|\theta(0)\rangle$ is orthogonal to all nonzero $|n\rangle$ and $\langle 0|\mathcal{V}|0\rangle=1$ we can write the normalized total dipole moment in the strong coupling limit, using Eqs. (23)–(26), in the form

$$\begin{aligned} \frac{M(t)}{M(0)} &= \left[1 - \epsilon^2 \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k^2} \right] \exp \left[- \left[1 - \epsilon \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k} \right] w_\rho t \right] \\ &+ \epsilon^2 \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k^2} \exp \left[- \left[E_k w_c + w_\rho \langle k|\mathcal{V}|k\rangle + \epsilon w_\rho \sum_{m(\neq k)} \frac{|\langle m|\mathcal{V}|k\rangle|^2}{E_k - E_m} \right] t \right]. \end{aligned} \quad (27)$$

correct to second order in the perturbation. As expected, for infinitely strong coupling between pores ($\epsilon=0$) the magnetization is single exponential with decay rate w_ρ reflecting the total surface to volume ratio in the system. The first correction to the decay rate (second order in the perturbation, first order in ϵ) decreases it below w_ρ . Note that the first correction to the amplitude is second order in the small parameter $\epsilon = w_\rho/w_c$. The multiple exponential spectrum due to second-order corrections consists of small amplitude contributions at very high rates. As $w_c \rightarrow \infty$, the dominant term in the decay rate is $E_k w_c$; thus we expect the decay rates to increase in proportion to w_c . These fast rates correspond to the transfer of magnetization between pores due to high coupling. From Eq. (17)

$$\begin{aligned} n(\lambda) &= \left[1 - \epsilon^2 \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k^2} \right] \delta \left[\lambda - \left[1 - \epsilon \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k} \right] w_\rho \right] \\ &+ \epsilon^2 \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k^2} \delta \left[\lambda - \left[E_k w_c + w_\rho \langle k|\mathcal{V}|k\rangle + \epsilon w_\rho \sum_{m(\neq k)} \frac{|\langle m|\mathcal{V}|k\rangle|^2}{E_k - E_m} \right] \right], \end{aligned} \quad (28)$$

a δ function of almost unit amplitude and small amplitude contributions at rates increasing in proportion to w_c . Note, to zeroth order in the perturbation, $\tau_m = \tau_0 = 1/w_\rho$ as expected [see Eq. (7)]. This perturbation expansion of

the spectrum of rates is valid when a magnetic moment has had time to explore enough of the pore space to be influenced by the average geometry of the system, as opposed to isolated sections, before decaying due to relaxa-

tion at the walls. In the next section, in connection with effective medium theory, we present a simple self-consistent perturbation theory (SCPT) which has some computational advantages.

We now turn to the weak coupling case. Here, the unperturbed Hamiltonian $\mathcal{H}^{(0)}$ is diagonal, so we find $E_n = r_n$ and $|n\rangle$ is a vector with 1 in the n th place and 0 elsewhere. The matrix elements $\langle k|\mathcal{V}|n\rangle$ can be evaluated analytically; we find

$$\langle k|\mathcal{V}|n\rangle = \sum_m \frac{s_{mn}}{v_n} \delta_{kn} - \frac{s_{kn}}{\sqrt{v_k v_n}}. \quad (29)$$

The total dipole moment to first order can be written explicitly in the weak coupling limit:

$$\frac{M(t)}{M(0)} = \sum_n \frac{v_n}{N} \left[1 - \frac{2\epsilon}{v_n} \sum_k \frac{s_{kn}}{r_n - r_k} \right] \times \exp \left[- \left[r_n w_\rho + w_c \sum_k \frac{s_{kn}}{v_n} \right] t \right]. \quad (30)$$

When the pores are decoupled ($\epsilon=0$, $w_c=0$) the total dipole moment contains a spectrum of decay rates corresponding to the spectrum of surface to volume ratios of the individual pores. The amplitudes of the exponential contributions are proportional to the volumes of the pores. The first-order correction for weak coupling changes the amplitudes up or down depending on the relative size of nearest-neighbor decay rates, and increases the decay rates by a factor proportional to the coupling rate w_c . The spectrum of rates is

$$n(\lambda) = \sum_n \frac{v_n}{N} \left[1 - \frac{2\epsilon}{v_n} \sum_k \frac{s_{k,n}}{r_n - r_k} \right] \times \delta \left[\lambda - \left[r_n w_\rho + w_c \sum_k \frac{s_{k,n}}{v_n} \right] \right]. \quad (31)$$

In this case, the zeroth order lifetimes τ_m and τ_0 are not equivalent: $\tau_m = \sum_n v_n / N w_\rho r_n$ and $\tau_0 = N / (w_\rho \sum_n v_n r_n)$. The former weights long times, the latter weights short times.

The perturbation theory expressions found for the total dipole moment and spectrum of rates in the strong and weak coupling limits are useful benchmarks for more elaborate calculations which allow us to explore the full range of coupling strength.

IV. EFFECTIVE MEDIUM THEORY

Effective medium theory (EMT) has been described previously for calculations of electrical conduction in resistor networks,⁹ as well as for electronic structure calculations where it is known as the coherent potential approximation (CPA).¹⁰ Random values of conductance are replaced by an average value, chosen such that when one resistor is changed back to its original value, the effect on the conductance is on average zero. EMT works best for very simply problems. Equation (8), the equation of motion for the dipole moment at site i , contains bond disorder, S_{ij} , as well as two kinds of site disorder, A_i and V_i .

To illustrate the application of EMT we examine the physical system in an extreme approximation. We take all of the cross-sectional areas equal and all of the volumes equal, i.e., $s_{ij}=1$ or 0 and $v_i=1$. We retain disorder only in the pore decay rates, e.g., due to variation of ρ from pore to pore ($r_i = \rho_i / \bar{\rho}$).

It is particularly convenient to work in the Laplace transform domain. Before we describe EMT, we review some exact relationships. From Eqs. (16) and (17) we have

$$\frac{\tilde{M}(s)}{M(0)} = \int_0^\infty \frac{n(\lambda)}{s+\lambda} d\lambda. \quad (32)$$

Since $\tilde{M}(s)$ is analytic everywhere except on the negative real axis, the contour integral which takes one from $\tilde{M}(s)$ to $M(t)$ can be wrapped around the negative real axis with the result that Eq. (3) is reproduced. Furthermore, $n(\lambda)$ can be obtained from Eq. (32) by

$$n(\lambda) = \frac{1}{\pi} \lim_{s \rightarrow -\lambda - i\epsilon} \text{Im} \left[\frac{\tilde{M}(s)}{M(0)} \right]. \quad (33)$$

If all the r_i are equal to a common value $\bar{r}$, then $n(\lambda) = \delta(\lambda - w_\rho \bar{r})$ and $\tilde{M}(s)/M(0) = [s + w_\rho \bar{r}]^{-1}$. When the values of r_i are not all equal, we define the complex valued function $\bar{r}(s)$ such that

$$\frac{\tilde{M}(s)}{M(0)} = \frac{1}{s + w_\rho \bar{r}(s)}. \quad (34)$$

The goal of EMT is to determine the optimal $\bar{r}(s)$, analogous to the self-energy in quantum mechanics (see below).

Following the EMT prescription we replace all decay rates r_i by an effective rate $\bar{r}$. We seek to find $\bar{r}$ such that changing the decay rate at one site, $i=0$, back to its original value does not change the total dipole moment, on average. Taking the Laplace transform of Eq. (8), using $r_i = \bar{r} + \Delta r \delta_{i0}$, where $\Delta r = r - \bar{r}$, we find

$$(s + w_\rho \bar{r}) \tilde{M}_i = \bar{m} \bar{V} - w_\rho \Delta r \tilde{M}_0 \delta_{i0} + w_c \sum_j^{\text{NN}} (\tilde{M}_j - \tilde{M}_i), \quad (35)$$

where $\bar{m}$ is the initial magnetization. We solve for the dipole moment on the zeroth site using a Fourier series transform and find

$$\tilde{M}_0 = \frac{\bar{m} \bar{V}}{s + w_\rho \bar{r}} \left[1 - \frac{w_\rho I(s, \bar{r}) \Delta r}{1 + w_\rho I(s, \bar{r}) \Delta r} \right], \quad (36)$$

where

$$I(s, \bar{r}) = \frac{1}{\pi^d} \int_0^\pi d\phi_1 \cdots \int_0^\pi d\phi_d \frac{1}{s + w_\rho \bar{r} + 4w_c \sum_{i=1}^d \sin^2 \frac{\phi_i}{2}}, \quad (37)$$

and d is the dimension of the hypercubic lattice. [We often use $I(s)$ as shorthand for $I(s, \bar{r})$ when there is no ambiguity.]

We impose the condition that at site zero the dipole moment averaged over the distribution of rates $p(r)$ is

the same as the total dipole moment of the model system, i.e., a system made up entirely of rates $\bar{r}$. This condition is imposed at finite s . Thus $\bar{r}$ is a function of s . We require

$$\int dr p(r) \frac{r - \bar{r}(s)}{1 + w_p I(s)[r - \bar{r}(s)]} = 0, \quad (38)$$

where $p(r)$ is the distribution of rates r . Equation (38) is a single-site approximation. Given a distribution of rates we can find an $\bar{r}$ for each s . Calculations of $n(\lambda)$, $M(t)$, τ_m , and τ_0 are straightforward, once we have $\bar{r}(s)$.

We note that if we write out the Hamiltonian in Eq. (10) with v_i and s_{ij} independent of i and j (the extreme approximation under discussion) we see that $\mathcal{H}_{ij}$ corresponds to a tight-binding Hamiltonian with site disorder r_i . Therefore, there is an isomorphism between the two problems:

$$\begin{aligned} s &\leftrightarrow z, \\ w_p r_i &\leftrightarrow \epsilon_i, \\ w_c &\leftrightarrow t, \\ I &\leftrightarrow F_0, \\ w_p \bar{r} &\leftrightarrow \Sigma, \\ n(\lambda) &\leftrightarrow A(\mathbf{k}=0, E). \end{aligned} \quad (39)$$

The quantities on the right are those defined in Ref. 11. Equation (38) is the CPA equation. Note in particular that the spectral density relevant to this problem, $n(\lambda)$, is not the total density of states, but rather it is the $\mathbf{k}=0$ component only, reflecting the condition that the system was initially magnetized uniformly.

A calculation of $n(\lambda)$ in EMT, requires a solution to the integral $I(s, \bar{r})$. Equation (37) can be solved in closed form in the case of a one-dimensional lattice,

$$I_1(s, \bar{r}) = \frac{1}{4w_c \sqrt{z} \sqrt{z+1}}, \quad (40)$$

$$I_3(s, \bar{r}) = \frac{1}{\pi^2 w_c} \int_0^{\pi/2} dx \int_0^{\pi/2} dy \frac{1}{\sqrt{z + \sin^2 x + \sin^2 y} \sqrt{z + \sin^2 x + \sin^2 y + 1}} \quad (41)$$

by brute force. Equation (37) can also be rewritten as a single integral in time over a modified Bessel function of complex argument raised to the dimensional power.¹¹ We did not pursue this method here.

Our first choice of decay rate distribution is the bimodal distribution given by

$$p(r) = \sin^2(\pi r), \quad 0 \leq r \leq 2. \quad (42)$$

In Figs. 3 and 4 we show $n(\lambda)$ calculated in EMT (the solid line) for the one-dimensional and three-dimensional cases, respectively. The spectra of decay rates were calculated for various values of w_c , $0.01 \leq w_c/w_p \leq 100$ in Fig. 3 and $0.01 \leq w_c/w_p \leq 0.2$ in Fig. 4. To find $n(\lambda)$ we

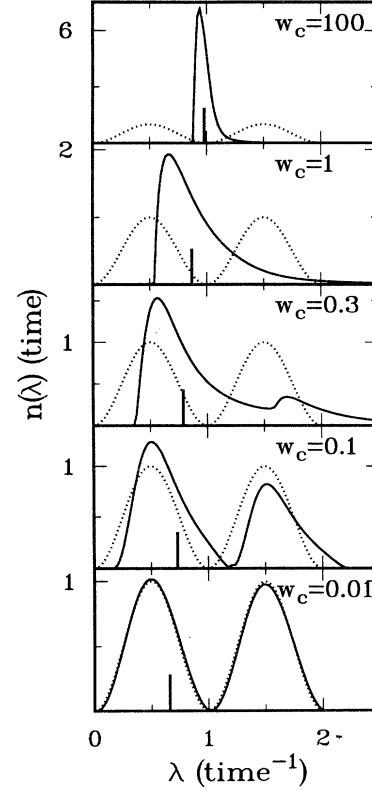


FIG. 3. One-dimensional spectra of decay rates from EMT. The distribution of individual pore decay rates is given by Eq. (42) (dotted lines). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by short vertical lines. As $w_c \rightarrow \infty$, $\tau_m \rightarrow \tau_0$.

where $z = (s + w_p \bar{r})/4w_c$. In higher dimension, the integral must be evaluated numerically. For the three-dimensional case, one integration can be done by hand and we did the remaining two-dimensional integral:

use the following procedure.

(1) Choose $w_p = 1$, $\epsilon = 0.001$ and $p(r)$ given in Eq. (42). We choose ϵ small enough that smaller choices produce no visible change in $n(\lambda)$.

(2) For each $s = -\lambda - i\epsilon$, numerically minimize the absolute value of the left-hand side of Eq. (38) with respect to $\bar{r}$.

(3) Use the resultant $\bar{r}(-\lambda - i\epsilon)$ and Eqs. (33) and (34) to find $n(\lambda)$.

The results shown in Figs. 3 and 4 confirm those expected from perturbation theory. In the weak coupling limit $n(\lambda)$ is the same as the original distribution of rates. In the strong coupling limit $n(\lambda)$ approaches a δ function

with a tail on the high rate side. Note that the rates in $n(\lambda)$ extend beyond the $\lambda=2$ limit of $p(r)$ as w_c increases. The mean lifetime τ_m , whose inverse is marked with a vertical line on the figures, approaches $\tau_0=1$ in the limit of infinite coupling. The most striking difference between the one- and three-dimensional cases is that the spectrum of decay rates narrows for much lower values of the coupling ratio in three dimensions than in one dimension. The degree of connectedness of the system, providing more ways for magnetic moments to move from one pore to another, has a strong effect on the resulting spectrum of decay rates.

Because of the connection with the quantum mechanical problem we can bring to bear on the problem set by Eqs. (9)–(17), any of a variety of approximate methods designed for electron and phonon problems. As an illustration we sketch in Appendix A the treatment of Eqs. (9)–(17) using the self-consistent perturbation theory (SCPT) formalism of Ehrenreich and Schwartz.¹⁰ The outcome of this treatment is a self-consistent equation that provides $\bar{r}$,

$$\bar{r} = \langle r \rangle - w_\rho \langle (r - \langle r \rangle)^2 \rangle I(s, \bar{r}(s)), \quad (43)$$

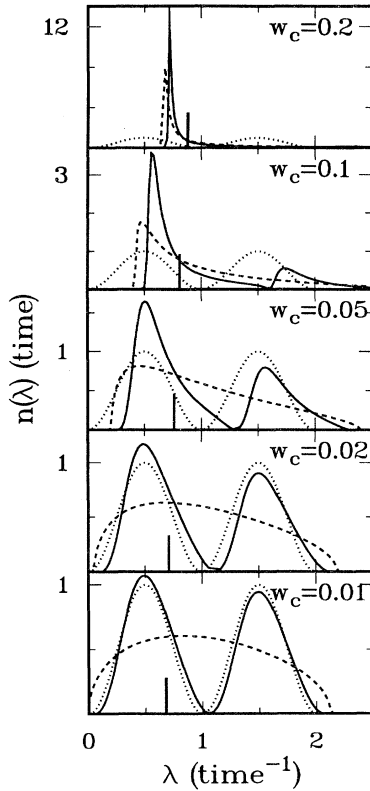


FIG. 4. Three-dimensional spectra of decay rates from EMT and SCPT. The EMT results are the solid lines; the SCPT results are the dashed lines. The distribution of individual pore decay rates is given by Eq. (42) (dotted lines). The coupling rate varies; $w_\rho = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} from EMT is marked by short vertical lines.

where $I(s, \bar{r}(s))$ is given by Eq. (37), for use in Eqs. (33) and (34). Equation (43) can also be found from approximate treatment of the EMT condition, Eq. (38). In Fig. 4 we show $n(\lambda)$ versus λ with $\bar{r}(s)$ from Eq. (43) (the dashed line) for the three-dimensional case. The distribution of decay rates is given by Eq. (42). Since Eq. (43) is dependent on the first and second moments of the rate distribution and an integral determined by the crystal structure only, SCPT captures the width of the spectrum of rates, but not the details of the exact $n(\lambda)$. The approximate treatment of EMT, Eq. (43), is in good agreement with its exact implementation for $\langle (\Delta r)^2 \rangle / w_c \ll 1$ ($w_c > 0.2$ in our example).

EMT is a valuable method for studying the spectrum of decay rates in a simple system. In this case, the system contains variation in pore values of ρ only. We are also interested in situations in which there are variations in pore radius and interpore throat size. Having more than one varying quantity in an EMT calculation leads to ambiguities in the choice of conditions and their meaning. Instead of expanding the discussion of EMT calculations we turn to matrix diagonalization.

V. MATRIX DIAGONALIZATION

In principle, the total dipole moment and spectrum of rates $n(\lambda)$ can be found exactly for any given porous system by diagonalizing the Hamiltonian in Eq. (10). In practice, computation time and memory constraints require the model porous medium systems we study by this method to be orders of magnitude smaller than any experimentally studied piece of material. However, for these small systems we may examine the consequences of variations in volume, surface area, and cross-sectional area.

We use the following procedure to find $n(\lambda)$.

(1) Diagonalize the Hamiltonian $\mathcal{H}_{ij}$ to find the eigenvalues and eigenvectors of the matrix.

(2) Find the overlap of each eigenvector $|\psi_n\rangle$ with $|\theta(0)\rangle$ given in Eq. (12). The square of this overlap is the amplitude of the spectrum of rates at the rate equal to the corresponding eigenvalue λ_n [see Eq. (17)].

In order to compensate for our finite-size systems we use periodic boundary conditions in the Hamiltonian and average $n(\lambda)$ over many realizations, typically 500 or 1000 for systems of $8 \times 8 \times 8 = 512$ pores. The results of matrix diagonalization (MD) in several cases are described in the following.

(1) The spectra of decay rates $n(\lambda)$ shown in Fig. 5 are for a one-dimensional lattice of pores. The pore volumes and throat cross-sectional areas are constant; the individual pore decay rates vary with the rate distribution given by Eq. (42). The lattice size is 512 pores and the results are averaged over 500 realizations. The coupling ratio w_c/w_ρ varies from 0.01 to 100; $w_\rho = 1$. The dashed curves are the EMT results shown in Fig. 3; we find the spectra of decay rates very well approximated by EMT. When $w_c = 10^{-2}$, the spectrum of decay rates is that of the individual pore decay rates. As w_c increases, the spectrum of decay rates collapses such that the majority of the amplitude is in a sharp peak at $\lambda = 1$. Although

high rate amplitudes are decreasing, the maximum rate is increasing proportional to w_c for large w_c/w_p as predicted from perturbation theory.

(2) In Fig. 6 we show $n(\lambda)$ for a three-dimensional lattice of pores. The pore volumes and throat cross-sectional areas are constant; the individual pore decay rates vary with the rate distribution given by Eq. (42). The lattice size is $8 \times 8 \times 8 = 512$ pores and the results are averaged over 500 realizations. The coupling ratio range is $0.01 \leq w_c/w_p \leq 0.2$; $w_p = 1$. Each pore has six nearest neighbors rather than two as in the one-dimensional case. This increased connectivity causes the spectrum of rates to collapse to a δ function much more rapidly with increasing w_c than for the one-dimensional case. The width of the spectrum of decay rates is already very narrow when $w_c = 0.2$, although $\tau_m^{-1} < w_p$. Although the EMT results of Fig. 4 are plotted in Fig. 6, they are almost not visible. The EMT approximation is known to improve as dimension is increased.

The previous examples have focused on cases where the decay rates vary between pores, but are independent of volume. In a physical system in the fast diffusion regime, $w_D/w_p > 1$, we expect an individual pore decay rate to be determined by the surface to volume ratio of the pore. In addition, the spectrum of decay rates measured

experimentally in uncoupled pores is the volume weighted distribution of individual pore decay rates. That is, larger pores will contribute more signal to a measurement since they contain more magnetic moments. Making the assumption that all pores are spherical we can assign volumes and decay rates to individual pores given a distribution of radii a_i ,

$$v_i = \frac{a_i^3}{\langle a^3 \rangle}, \quad r_i = \frac{1}{a_i} \frac{\langle a^3 \rangle}{\langle a^2 \rangle}, \quad (44)$$

where $\langle a^n \rangle$ is the average of a_i^n . Allowing v_i to vary introduces substantial variation in the coupling terms of the Hamiltonian $\mathcal{H}$, even if there is no variation in cross-sectional area s_{ij} .

(3) In Fig. 7 we show $n(\lambda)$ for a three-dimensional lattice of pores. The throat cross-sectional areas are constant; the pore radii are chosen from the distribution

$$p(a) = \frac{A}{a^5} \sin^2 \left[\frac{2\pi a_{\min}}{a} \right], \quad a_{\min} \leq r < \infty, \quad (45)$$

where $a_{\min}$ and A are chosen such that $\langle a \rangle = 1$ and the spectrum of rates $n(\lambda)$ is given by Eq. (42) when the pores are uncoupled. The lattice size is $8 \times 8 \times 8 = 512$ pores and the results are averaged over 1000 realizations.

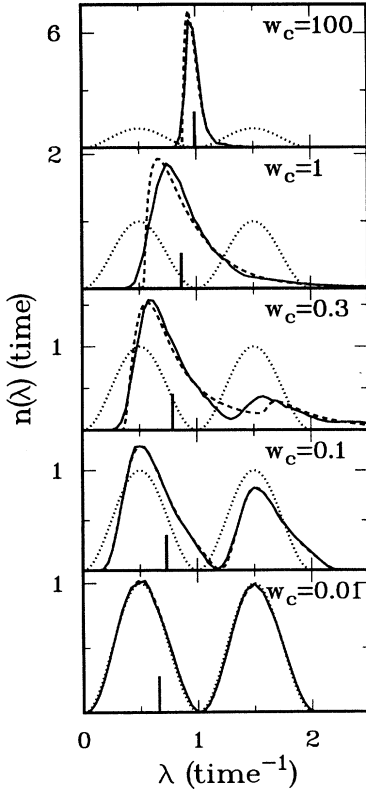


FIG. 5. One-dimensional spectra of decay rates from MD. The distribution of pore decay rates is given by Eq. (42) (dotted line). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by vertical lines. The EMT results of Fig. 3 are the dashed lines.

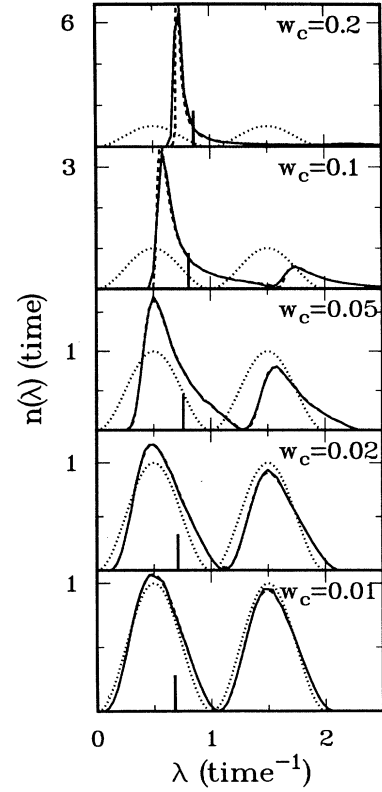


FIG. 6. Three-dimensional spectra of decay rates from MD. The distribution of pore decay rates is given by Eq. (42) (dotted line). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by vertical lines. The dashed lines are the EMT results of Fig. 4.

The coupling ratio range is $10^{-2} \leq w_c/w_p \leq 10^2$; $w_p=1$. Note that the distribution of pore radii is heavily weighted toward small pores. In order that the volume weighted distribution of rates at zero coupling have equal size peaks above and below $\lambda=1$, 95% of the pores have decay rates in the range $1 < \lambda < 2$. Thus we can think of the large pores (small decay rates) as dilute disorder in a system of very small pores. Initially, as w_c is increased, the small pores couple to each other forming a narrow peak in the decay spectrum at $\lambda \approx 1.5$, while the large pores remain relatively unaffected. This is due to the comparatively large number of small pores and also to the inverse volume effect in the coupling terms of the Hamiltonian [see Eq. (10)]. The effective coupling of large pores to their neighbors is much smaller than that of small pores. This feature is retained even to $w_c=10^2$; $n(\lambda)$ still has finite amplitude at small λ in contrast to the results of Fig. 6.

(4) In Fig. 8 we show $n(\lambda)$ for a three-dimensional lattice of pores. The throat cross-sectional areas are constant; the pore radii are chosen from a log-normal distribution

$$p(a) = \frac{1}{\sigma a \sqrt{2\pi}} \exp \left[-\frac{(\ln a - u_0)^2}{2\sigma^2} \right], \quad (46)$$

where $u_0 = -\frac{1}{4}$ and $\sigma^2 = \frac{1}{2}$, values which make the average pore radius one. Experimental measurements on rocks show the pore size distribution may vary over an order of magnitude or more.³ The distribution of Eq. (46) is an attempt to mimic a very broad spectrum such as might be seen in rocks. The lattice size $8 \times 8 \times 8 = 512$ pores and the results are averaged over 1000 realizations. The coupling ratio range is $0 \leq w_c/w_p \leq 100$; $w_p=1$. The raggedness of the results decreases slowly with increased number of realizations and increased bin size; each smooth curve is constructed using 50 bins spaced evenly on a logarithmic scale from 10^{-1} to 10^1 . The small λ part of the spectrum has substantial amplitude even for large coupling rates, due to the inverse volume effect in the coupling term of the Hamiltonian. Decay due to coupling is enhanced for small pores and decreased for large pores when volume effects are included in the coupling term of the Hamiltonian.

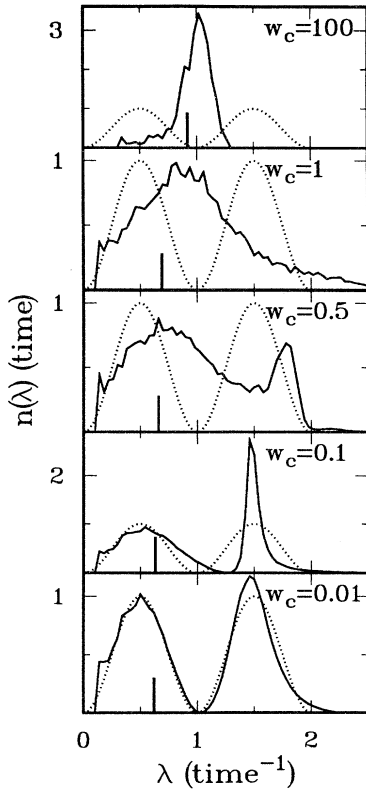


FIG. 7. Spectra of decay rates from MD when rates and volumes are correlated. The distribution of pore radii is given by Eq. (45). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by vertical lines. The dotted lines are the uncoupled pore result, Eq. (42).

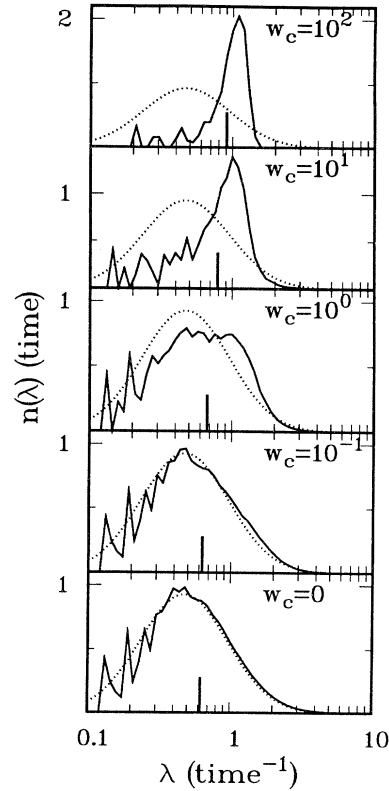


FIG. 8. Spectra of decay rates from MD for a broad distribution of pore radii. The distribution of pore radii is given by Eq. (46). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by vertical lines. The dashed lines are $n(\lambda)$ for the uncoupled system.

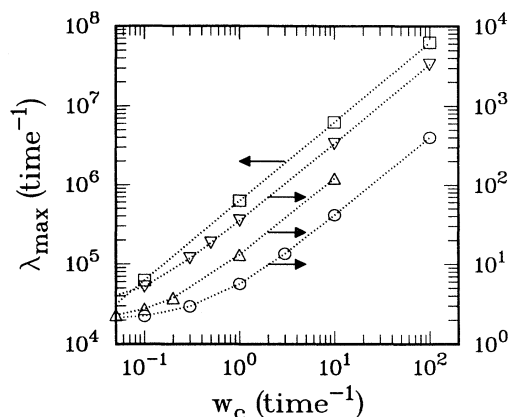


FIG. 9. Maximum decay rate as a function of coupling rate. The maximum decay rates are shown for the MD results of Figs. 5 (○) and 6 (△) for which the distribution of pore decay rates is given by Eq. (42), and Figs. 7 (□) and 8 (▽) for which the distributions of pore radii are given by Eqs. (45) and (46), respectively. $\lambda_{\max} \propto w_c$ as predicted by perturbation theory in the strong coupling limit.

One of the predictions of perturbation theory in the strong coupling limit [see Eq. (28)] is that the small amplitude contributions to $n(\lambda)$ are at rates which increase in proportion to the coupling rate; $\lambda \propto w_c$. We show this to be the case in Fig. 9, where the maximum decay rate $\lambda_{\max}$ is plotted against the coupling rate w_c for all four of the MD examples. Increased coupling causes very high frequency modes of very low amplitude as the magnetization flux from one pore to its neighbors increases. For the special cases in which only the pore rates vary, it is simple to show from Eq. (19) that $\lim_{w_c \rightarrow \infty} \lambda_{\max} = 4dw_c$, where d is the dimension. This is borne out in Fig. 9.

MD is a powerful method for studying the magnetization equation for connected pore systems. Any quantity appearing in the equation may be varied. We chose to study systems where only the individual pore rate varied, confirming our EMT results, and where both rate and volume vary. There are many more options to consider: the throat cross-sectional areas between pores are not all the same in a physical system, and the surface area and volume of a pore are not necessarily strictly related. MD is the most accurate method we discuss for studying the effects of a variety of geometrical features on the total dipole moment spectrum of decay rates.

VI. CONCLUSIONS

We have developed equations governing the magnetization evolution in fluid-filled connected pore systems. The evolution of the total dipole moment can be described by a spectrum of decay rates which reflects the decay rates of individual pores as well as the strength of the coupling between pores. We have used three methods, perturbation theory, effective-medium theory, and matrix diagonalization, to study the evolution of a

particular system's decay-rate spectrum as the coupling between pores is increased. The results of all three methods are in good agreement. The important qualitative features appearing in all three methods of solution are (1) when the coupling between pores is zero or very weak, the spectrum of decay rates is that of the individual pores. (2) As the coupling is increased, the range of decay rates increases, shifting to high rates. (3) For very strong coupling, the spectrum of decay rates narrows to a peak at a rate determined by the ratio of the total surface to total volume in the pore system.

The mean lifetime τ_m is a single parameter which characterizes a spectrum of decay rates in a model independent way. In Fig. 10 we summarize our results for τ_m as a function of the coupling rate w_c for the various situations we have encountered. In all cases, τ_0 is unity. The EMT and MD results for similar distribution and dimension are in good agreement. The upper pair of curves are from the three-dimensional MD calculations using log-normal [Eq. (46)] and bimodal [Eq. (45)] distribution of pore sizes. The central pair of curves is from the one-dimensional MD and EMT calculations using the bimodal distribution of pore decay rates given by Eq. (42). The lower triplet is from the three-dimensional MD, EMT, and SCPT calculations using the bimodal distribution of pore decay rates given by Eq. (42). The most notable feature of this figure is that the crossover from uncoupled to strongly coupled behavior occurs over approximately two decades in the coupling strength.

In order to relate our theoretical study to physical systems we must make some estimate of the coupling between pores. Two examples of porous media with widely different properties are sandstone and sol-gel glass. Both

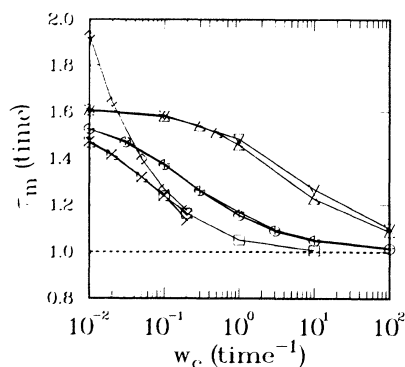


FIG. 10. Mean lifetime versus coupling strength. In the strong coupling limit, $\tau_m \rightarrow \tau_0 = 1.0$. The lines are a guide for the eye. The mean lifetime is plotted for all the cases previously discussed. For distribution of pore decay rates given by Eq. (42): the one-dimensional cases in Figs. 3 (+) and 5 (○), and the three-dimensional cases in Fig. 4 (×, EMT; ◇, SCPT), and 6 (□). For the three-dimensional cases using the distribution of pore radii given in Eq. (45) and shown in Fig. 7 (△), and using the distribution of pore radii given in Eq. (46) and shown in Fig. 8 (▽).

systems have been studied extensively using nuclear magnetic resonance to measure the decay of the total dipole moment in water-saturated samples. Using data for various sandstones from Kenyon *et al.*,³ we estimate a typical porosity of 20%, pore diameter of 80 μm , and throat diameter of 30 μm . A typical mean T_1 is 250 ms; we take this to be approximately τ_0 . The data were taken at room temperature, where the diffusion coefficient of water is $D = 2.5 \times 10^{-5} \text{ cm}^2/\text{s}$. These numbers imply a coupling strength $w_c/w_p < 0.1$. Kenyon *et al.* have decomposed their total dipole moment measurements into spectra of decay times which are one or more orders of magnitude wide. The decomposition is dependent on the assumption that the pores are uncoupled, i.e., $w_c \approx 0$. Given our small estimated coupling strength and the very broad distribution of pore sizes typically found in sandstones, the assumption that a spectrum of decay rates reflects the density of pore sizes in the rocks is probably a good one. Data on sandstones correspond to the $w_c \leq 0.1$ spectra in Fig. 8.

The sol-gel glasses characterized by Bhattacharja, *et al.*¹² have very different characteristics from the sandstones just discussed. The typical porosity in these systems is 85%, a representative pore diameter (assuming spherical pores) is 0.50 μm , and an average throat diameter is 0.24 μm . The magnetization decay measured for these water-saturated silica glasses was single exponential with a decay time $T_1 \approx 100$ ms at room temperature. These numbers imply a coupling strength $w_c/w_p = 10^3$, which is certainly in agreement with the single exponential result. In such highly coupled pore systems, we would expect the spectrum of decay rates to be a δ function, particularly in sol-gel glasses where the distribution of pore sizes is narrow.

An obvious question is whether there exist physical systems in which a range of coupling strength between pores can be explored. A possible method of attack is to change the diffusion coefficient of the saturating fluid by raising or lowering the temperature. In this way the coupling strength between pores may be changed by an order of magnitude or more. On the other hand, if the porous sample used in such an experiment has a very broad spectrum of pore sizes, the extra range in coupling may not make an experimentally evident change in the spectrum of decay rates. In the future we hope to compare our results with relevant experiments exploring coupling between pores.

ACKNOWLEDGMENTS

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APPENDIX: SELF-CONSISTENT PERTURBATION THEORY IN THE STRONG COUPLING LIMIT

Here we employ a perturbation theory in the strong coupling limit appropriate to the problem in which there is a decay rate distribution whose width is small compared to the coupling parameter w_c . We write the Hamiltonian as the sum of an unperturbed term describing a perfect tight-binding crystal and a perturbation term due to small fluctuations in rates:

$$\mathcal{H} = \mathcal{H}^{(0)} + \mathcal{V}, \quad (\text{A1})$$

where

$$\mathcal{H}_{ij}^{(0)} = (w_p r^{(0)} + x w_c) \delta_{ij} - w_c \delta_{ij_{nn}} \quad (\text{A2})$$

and

$$\mathcal{V}_{ij} = \Delta r_i \delta_{ij}. \quad (\text{A3})$$

Here x is the number of nearest neighbors, j_{nn} is a nearest neighbor to i , $r^{(0)}$ is an arbitrary dimensionless rate describing the perfect crystal, and $\Delta r_i = r_i - r^{(0)}$. The average Green function defines a self-energy, Σ , through the expression

$$\langle G \rangle = \left\langle \frac{1}{z - \mathcal{H}^{(0)} - \mathcal{V}} \right\rangle \equiv \frac{1}{z - \mathcal{H}^{(0)} - \Sigma}. \quad (\text{A4})$$

Expanding Σ to second order in $\mathcal{V}$ we find

$$\Sigma \approx \langle \mathcal{V} \rangle + \langle \mathcal{V} G^{(0)} \rangle - \langle \mathcal{V} \rangle G^{(0)} \langle \mathcal{V} \rangle, \quad (\text{A5})$$

where $G^{(0)}$ is the Green function for a perfect tight-binding crystal. In the magnetization problem, the self-energy describes the change in an effective rate of decay from the reference point $r^{(0)}$. Thus, inserting the perfect crystal Green function, we find the $\bar{r}$ of Eq. (34) can be written

$$\bar{r} = \langle r \rangle - w_p \langle (r - \langle r \rangle)^2 \rangle I(s, r^{(0)}), \quad (\text{A6})$$

where $I(s, r^{(0)})$ is given by Eq. (37). The method of SCPT suggests we replace $r^{(0)}$ in the integral $I(s, r^{(0)})$ with $\bar{r}$. We then have a self-consistent equation for $\bar{r}$ which is independent of the reference value $r^{(0)}$ in the original problem [Eq. (43)]. In this formalism the value of $\bar{r}$ is given by the average and the second moment of the distribution of rates and an integral which depends only on the crystal structure. It is straightforward, but slightly tedious, to show that Eqs. (43) and (28) are equivalent to first order in ϵ .

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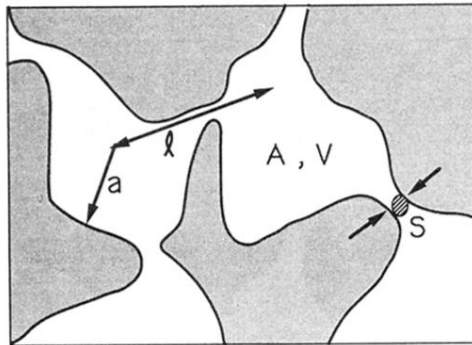


FIG. 1. Schematic of a porous-medium cross-sectional area. The porous medium is characterized by large voids, each assigned a volume V and surface area A , connected by narrow channels of cross-sectional area S . The distance between pores is l and $a = V/A$.

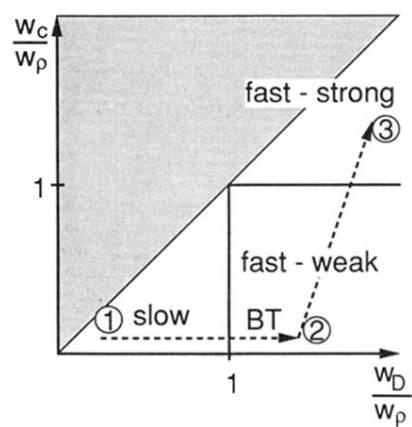


FIG. 2. Rate space for magnetization decay in pore systems. Brownstein and Tarr discuss evolution from slow to fast diffusion, $1 \rightarrow 2$. We discuss evolution from weak to strong coupling, $2 \rightarrow 3$.